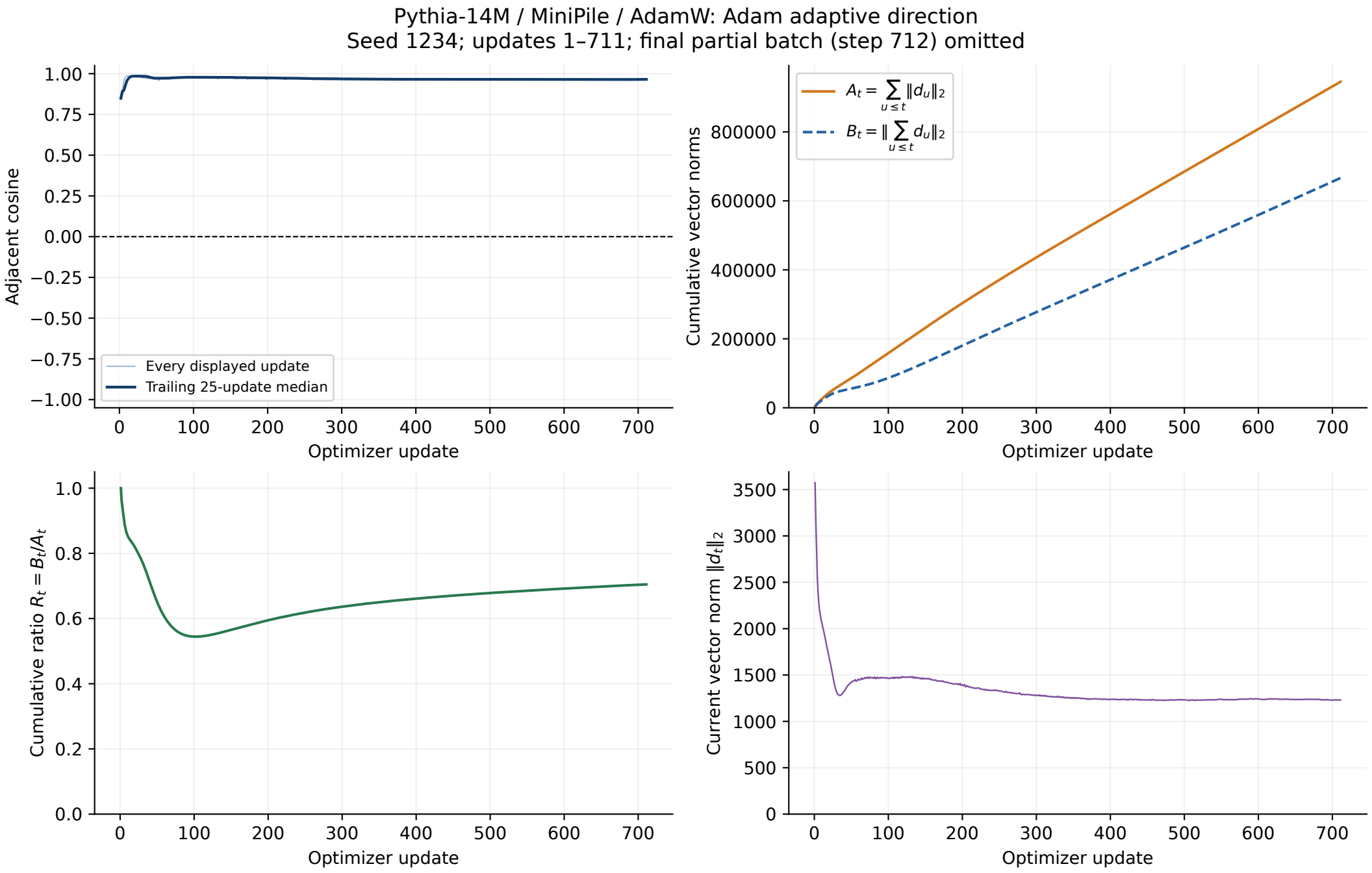

Pythia-14M / MiniPile / AdamW: Adam adaptive direction
Seed 1234; updates 1–711; final partial batch (step 712) omitted
Adjacent cosine
Every displayed update
Trailing 25-update median
Optimizer update
Cumulative vector norms
$A_t = \sum_{u \le t} \|d_u\|_2$
$B_t = \| \sum_{u \le t} d_u \|_2$
Optimizer update
Cumulative ratio $R_t = B_t / A_t$
Optimizer update
Current vector norm $\|d_t\|_2$
Optimizer update